

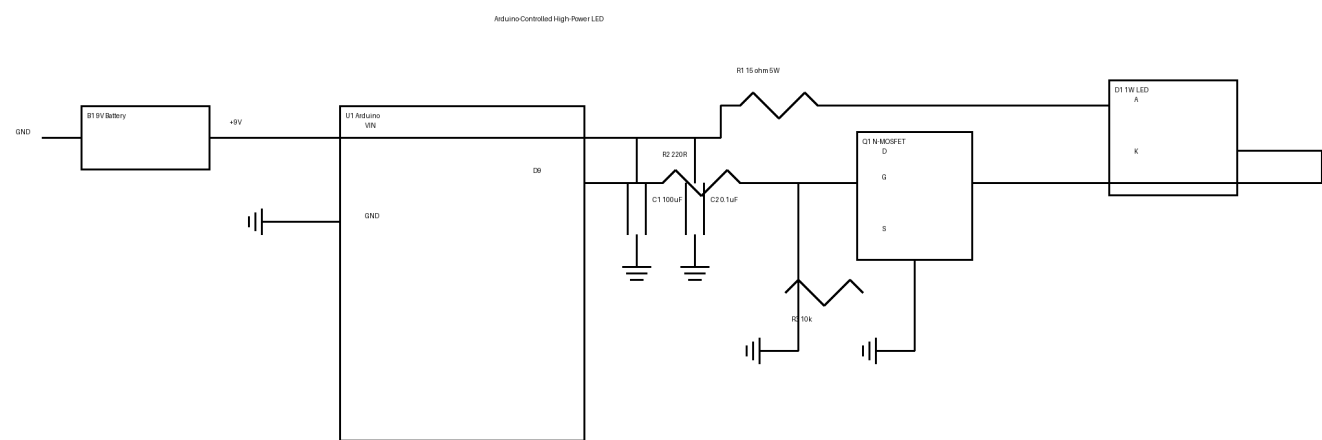
Arduino-Controlled High-Power LED from 9 V Battery

Goal: switch a high-power LED from a 9 V source using an Arduino and a MOSFET driver stage.

Assumptions

- Battery source is a 9 V DC battery or equivalent 9 V supply.
- High-power LED assumed to be a 1 W class white LED around 3.2 V forward voltage at about 350 mA.
- Arduino is assumed to be an Arduino Uno/Nano class 5 V board.
- Because a high-power LED should not be driven directly from an Arduino pin, the LED is switched with a logic-level N-channel MOSFET.
- LED current is limited with a dedicated series resistor for a simple buildable design; a constant-current driver would be a better production choice.
- Arduino is powered from the same 9 V source through VIN/barrel input for conceptual simplicity.

Schematic



Common ground required between battery, Arduino, and MOSFET source

Bill of Materials

RefDes	Description	Qty	Unit \$	Ext. \$	Supplier / Link
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B1	9 V battery or regulated 9 V DC source	1	4.00	4.00	Generic https://www.amazon.com/s?k=9v+battery
U1	Arduino Uno or Nano compatible 5 V board	1	8.00	8.00	Arduino-compatible https://www.amazon.com/s?k=arduino+uno+compatible
D1	High-power white LED, ~1 W, $V_f \sim 3.2$ V @ 350 mA	1	1.50	1.50	Star-mounted LED https://www.amazon.com/s?k=1w+white+led+star
Q1	Logic-level N-channel MOSFET, e.g. IRLZ44N	1	1.20	1.20	IRLZ44N https://www.digikey.com/en/products/filter/mosfets-single/278
R1	15 ohm, 5 W resistor for approximate LED current limiting	1	1.50	1.50	Wirewound resistor https://www.digikey.com/en/products/filter/chassis-mount-resistors/54
R2	220 ohm resistor, MOSFET gate series resistor	1	0.05	0.05	Generic https://www.digikey.com/en/products/filter/through-hole-resistors/53
R3	10 kohm resistor, MOSFET gate pulldown	1	0.05	0.05	Generic https://www.digikey.com/en/products/filter/through-hole-resistors/53
C1	100 uF electrolytic capacitor, ≥ 16 V	1	0.20	0.20	Generic https://www.digikey.com/en/products/filter/aluminum-electrolytic-capacitors/58
C2	0.1 uF ceramic capacitor	1	0.05	0.05	Generic https://www.digikey.com/en/products/filter/ceramic-capacitors/60
J1	Battery clip / wiring leads	1	0.80	0.80	Generic https://www.amazon.com/s?k=9v+battery+clip
Total estimated cost				17.35	

Firmware file: high_power_led_arduino_9v.ino

Safety note: this simple resistor-limited design is conceptually buildable but a constant-current LED driver is the better engineering choice for sustained high-power LED operation.